Course Code: CS2226 **Name of the Programme:** B. Tech (H) **Name of the Course:** Machine Learning for Data science **Semester:** IV		
Course credit	**No. of hours per week (L + T + P)**	**Total no. of Teaching hours**
3	(1:1:2)	60

Course Objectives:

a) Introduce the fundamental principles of neural networks and their role in the machine learning and data science ecosystem.

b) Provide hands-on experience in building and training convolutional neural networks and sequences learning model using various modern architectures and techniques.

c) Explore unsupervised learning approaches and generative models such as autoencoders and GANs for data representation and synthesis.

d) Enable students to understand and apply model compression, optimization, and deployment strategies on edge devices and cloud platforms

Course outline

Syllabus:

Module – 1 Machine learning Foundations for Data Science Applications	**No. of hours**
	3
Machine learning and deep learning in solving complex data science problems. Types of Machine Learning, Overfitting and Underfitting, Understand the architecture of shallow and deep neural networks, activation functions, and training mechanisms through forward and backward propagation. Explore common loss functions like Cross-Entropy and MSE, and various optimization techniques. Examine weight initialization strategies essential for robust model convergence in real-world datasets.	

Module – 2 Deep Learning for Visual Data Analytics	**No. of hours**
	3
Analyse image data using Convolutional Neural Networks (CNNs), a key tool in computer vision for data science tasks such as classification and object detection. Study CNN layers, pooling, and activation strategies. Delve into advanced convolution	

techniques—Depthwise Separable, Dilated, and Pointwise convolutions—and regularization methods like Batch Normalization and Dropout. Practice using pretrained models (e.g., VGGNet, ResNet, EfficientNet, DenseNet, Mobile net) and apply transfer learning, fine-tuning, and feature extraction

Module – 3 Sequential Data Analysis	**No. of hours**
	3
Modelling time-dependent or sequential data, such as stock prices, user behaviour, and natural language. Fundamentals of recurrent neural networks (RNNs)and limitations. LSTM and GRU architectures for improved sequence retention and learn their real-world applications in forecasting and NLP. Transformer architecture for data science applications.	

Module – 4 Unsupervised Learning for Feature Extraction and Data Generation	**No. of hours**
	3
Unsupervised learning techniques used in dimensionality reduction, and synthetic data generation for Data science applications. Unsupervised Learning and Generative Modelling with Autoencoders and GANs: Autoencoder (Types: Deep, CNN-based), Training Autoencoders, Variational Autoencoders (VAE), Introduction to GAN, GAN Architecture (Generator, Discriminator)	

Module – 5 Model Optimization and Deployment in the Data Science Pipeline	**No. of hours**
	3
Model compression techniques: Pruning, Quantization, Knowledge Distillation, ONNX and model conversion formats for deployment, Deep learning deployment on edge: NVIDIA TensorRT, OpenVINO, TFLite, Containerization and deployment: Docker, Flask/FastAPI, Streamlit, Deployment on cloud platforms: AWS SageMaker, GCP AI Platform, Azure ML.	

Course outcomes
a) Apply basic concepts of neural networks and deep learning in solving real time problems
b) Build deep learning models using convolutional and sequence learning techniques to process image, time-series, and text data.
c) Develop unsupervised learning models for Image Denoising, Image generation, anomaly detection.

d) Practices techniques for optimizing and deploying machine learning models in real-world environments.

Text Books	
1	Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. Deep learning. MIT Press, 2016, ISBN: 9780262035613.
2	Zhang, Aston, et al. "Dive into deep learning." Cambridge University Press, 2023, ISBN-13: 9781009389433.

Reference books	
1	Michael Nielsen. Neural Networks and Deep Learning. 2019. Online book.
2	Aurélien Géron. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. Shroff/O'Reilly. Third edition. ISBN-10: 1492032646
3	CS6910: Deep Learning, IIT Madras.
4	CS231n: Deep Learning for Computer Vision, Stanford University.
5	CS 4263/5483: Deep Learning, University of Texas at San Antonio
6	(6.S191) Introduction to Deep Learning, MIT

Tutorials	**[15 Hours]**
1	Linear Algebra for Neural Networks
2	Calculus for Deep Learning – Derivatives and Backpropagation
3	Optimization Techniques in Deep Learning
4	Mathematical Foundations of CNNs
5	Mathematics of Recurrent Networks and LSTMs
6	Generative Models and Latent Space Mathematics

Lab Programs /Tutorials	**[30 Hours]**
1	Building and Training a Basic Neural Network for Binary Classification
2	Implementing Forward and Backward Propagation from Scratch for Deep neural network
3	Designing and Training a Simple Convolutional Neural Network
4	Comparative Study of Modern CNN Architectures Using Transfer Learning
5	Developing Sequence Models Using RNN
6	Developing Sequence Models Using LSTM, and GRU
7	Building Deep and CNN based Autoencoders
8	Building Variational Autoencoders
9	Generating Synthetic Images Using GANs
10	Model compressions and deployment

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Lesson plan (Template –2)

Name of the Programme:	**B. Tech (Hons.)**	
Title of the Course:	**Machine Learning for Data science**	
Course code	**CS2226**	
Semester:	**IV**	
Names of the Course Instructor(s)	**Dr. P. Karthikeyan, Prof: Ashwini Kumar Mathur**	
Course credit	**No. of hours per week (1+1+2)**	**Total no. of Teaching hours**
03	**04 Hours**	**60 Hours**

Session No.	Module	Hours	Topic	Pedagogy / Activities	Pre-reading / Reference
1	1	1	ML & DL in Data Science	Interactive lecture, real-world examples	Goodfellow et al., Ch.1
2	1	1	Types of ML, Overfitting & Underfitting	Concept discussion, bias–variance analysis	Géron, Ch.2
3	1	1	Neural Network Architecture & Training	Forward/backpropagation derivation	Nielsen, Ch.2
4	1	1	Linear Algebra for Neural Networks	Problem-solving session	D2L – Math Appendix
5	1	1	Calculus for Backpropagation	Gradient derivations	Nielsen, Ch.2
6	1	1	Optimization Techniques	SGD vs Adam comparison	Géron, Ch.11
7	1	1	Basic Neural Network	Binary classification lab	TensorFlow Docs
8	1	1	Forward Propagation from Scratch	Hands-on implementation	D2L Ch.3
9	1	1	Backward Propagation from Scratch	Gradient computation	D2L Ch.3
10	2	1	CNN Fundamentals	Visual demos, convolution intuition	CS231n
11	2	1	CNN Layers & Pooling	Layer-wise explanation	Géron, Ch.14
12	2	1	Advanced CNNs & Transfer Learning	VGG, ResNet discussion	MIT 6.S191
13	2	1	Mathematics of CNNs	Kernel & feature map analysis	CS231n
14	2	1	Regularization Techniques	Dropout, BatchNorm analysis	Goodfellow, Ch.8

15	2	1	CNN Architecture Comparison	Group discussion	D2L
16	2	1	Simple CNN Design	CNN implementation	Keras Docs
17	2	1	CNN Training & Evaluation	Model tuning	TensorFlow
18	2	1	Transfer Learning – VGG	Feature extraction	CS231n
19	2	1	Transfer Learning – ResNet	Fine-tuning	ResNet Paper
20	2	1	MobileNet / EfficientNet	Lightweight models	MobileNet
21	2	1	CNN Performance Comparison	Accuracy vs speed analysis	D2L
22	2	1	Image Classification Project	Mini-project	Instructor Notes
23	2	1	Results & Report Writing	Interpretation	—
24	3	1	Sequential Data & RNNs	Time-series modelling	Géron, Ch.15
25	3	1	LSTM & GRU Architectures	Gate mechanisms	D2L Ch.9
26	3	1	Transformer Architecture	Attention mechanism	Vaswani et al.
27	3	1	RNN Mathematics	Unrolling sequences	D2L
28	3	1	LSTM/GRU Mathematics	Gate equations	MIT 6.S191
29	3	1	Sequential Use Cases	Forecasting & NLP	Géron
30	3	1	RNN for Time Series	Hands-on lab	TensorFlow
31	3	1	RNN Optimization	Hyperparameter tuning	Keras
32	3	1	LSTM Implementation	Sequence learning	TensorFlow
33	3	1	GRU Implementation	Model comparison	TensorFlow
34	3	1	Forecasting / NLP Task	Applied experiment	D2L
35	3	1	Evaluation of Sequence Models	Metrics analysis	—
36	4	1	Unsupervised Learning Concepts	Dimensionality reduction	Géron, Ch.8
37	4	1	Autoencoders & VAEs	Latent space intuition	D2L Ch.17
38	4	1	GAN Architecture	Generator–Discriminator	Goodfellow, Ch.20
39	4	1	Latent Space Mathematics	KL divergence	Goodfellow
40	4	1	Reconstruction Loss	Mathematical analysis	D2L

41	4	1	Generative Model Comparison	Discussion	MIT 6.S191
42	4	1	Deep Autoencoder	Feature extraction	TensorFlow
43	4	1	CNN Autoencoder	Image denoising	Keras
44	4	1	Variational Autoencoder	Probabilistic modelling	D2L
45	4	1	GAN Implementation	Image generation	TensorFlow
46	4	1	GAN Training Stability	Mode collapse study	Goodfellow
47	4	1	Generated Data Evaluation	Visual assessment	—
48	5	1	Model Compression Techniques	Pruning & quantization	Géron, Ch.19
49	5	1	Edge Deployment	TensorRT, TFLite	NVIDIA Docs
50	5	1	Cloud Deployment	AWS, GCP, Azure	Cloud Docs
51	5	1	ONNX & Model Conversion	Workflow explanation	ONNX Docs
52	5	1	API-based Deployment	Flask vs FastAPI	FastAPI
53	5	1	Deployment Case Studies	Industry discussion	Blogs
54	5	1	Model Pruning & Quantization	Hands-on	TensorFlow
55	5	1	ONNX Conversion	Model export	ONNX
56	5	1	Edge Deployment	TFLite/TensorRT	NVIDIA
57	5	1	Containerization	Docker	Docker Docs
58	5	1	Cloud Deployment	AWS/GCP demo	Cloud Console
59	5	1	End-to-End ML Pipeline	Complete workflow	Instructor Notes
60	5	1	Final Review & Assessment	Model evaluation	—

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 20 Marks QUIZ	20 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Lab record & viva - Midterm Project Review	 5 marks out of 70 5 marks out of 70

- Final Project Submission	15 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Practical
Weightage of exam	30%

Course Outcomes- Progamme Outcomes Matrix

Program Outcome / Course Outcome		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	Apply basic concepts of neural networks and deep learning in solving real time problems	3	2	3		3	1						
CO2	Build deep learning models using convolutional and sequence learning techniques to process image, time-series, and text data.	3	2	3		3							
CO3	Develop unsupervised learning models for Image Denoising, Image generation, anomaly detection.	3	2	3	2	2	1						
CO4	d) Practices techniques for optimizing and deploying machine learning models in real-world environments.	3	2	3	2	2	1						

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		**20**
CIE-1 Components		**Rubrics for Assessment**
CIE-1	Quiz Graded Component 1	Quiz will be conducted for 20 marks. Each right answer carries one mark each. No negative marking for wrong answers.

Total Marks		**25**
CIE-2 Components		**Rubrics for Assessment**
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.

Total Marks		**25**
CIE-3 Components		**Rubrics for Assessment**
CIE-3	Practical Component	Lab record+ Viva = 5 marks (*Refer Table-1)* Midterm Project review = 5 Marks (*Refer Table 3)* Final Project Submission = 15 marks *(Refer Table-2)*

Table 1: Lab Write-up, Execution Rubrics and Viva Voce Rubrics (Maximum: 10 Marks)

Lab Write-up and Execution rubrics (Max: 6 marks)					
S. No	**Criteria**	**Measuring Methods**	**Excellent**	**Good**	**Poor**
1	**Understanding of problem statement.** **(2 Marks)**	Observations	Student exhibits thorough understanding of requirements and applies suitable algorithm for the problem (**2 M)**	Student has sufficient understanding of requirements and applies suitable algorithm for the problem. **(1 M)**	Student does not have a clear understanding of requirements and is unable to apply suitable algorithm for the problem. **(0 M)**

2	**Execution** **(2 Marks)**	Observations	Student demonstrates the execution of the program with optimized code and shows performance efficiency. **(2 M)**	Student demonstrates the execution of the program without optimization of the code and shows performance efficiency. **(1 M)**	Student has not executed the program. **(0 M)**
3	**Results and Documentation (2 Marks)**	Observations	Documentation with appropriate comments and output is covered in data sheets and manual. **(2 M)**	Documentation with only few comments and only few output cases is covered in data sheets and manual. **(1 M)**	Documentation with no comments and no output cases is covered in data sheets and manual. **(0 M)**
			Viva Voce Rubrics (Max: 4 marks)		
1	**Conceptual Understanding (2 Marks)**	Viva Voce	Explains thoroughly the ML algorithm and related concepts. **(2 M)**	Adequately explains the ML algorithm and related concepts. **(1 M)**	Unable to explain ML algorithm and related concepts. **(0 M)**
2	**Use of appropriate Problem-Solving Techniques (1 Marks)**	Viva Voce	Insightful explanation of appropriate problem-solving techniques for the given problem to derive solution. **(1 M)**	Sufficiently explains the use of appropriate problem-solving techniques for the given problem to derive solution. **(0.5 M)**	Unable to explain the problem-solving techniques for the given problem. **(0 M)**
3	**Communication of Concepts (1 Marks)**	Viva Voce	Communicates the concept used in problem solving well. **(1 M)**	Sufficiently communicates the concepts used in problem solving. **(0.5 M)**	Unable to communicate the concepts used in problem. **(0 M)**

Table 2: Project Components Evaluation Rubric (Maximum: 15 Marks)

Project Components **[15 Marks]**	**Excellent**	**Very Good**	**Good**	**Not yet completed**
Presentation **[4 Marks]**	Engaging, clear, and well-structured presentation that effectively communicates key points. **4 Marks**	Presentation is mostly clear and organized, with minor areas for improvement. **3 Marks**	Presentation is somewhat unclear or disorganized, with noticeable room for improvement. **2 Marks**	Presentation is missing or significantly incomplete. **0-1 Marks**
Implementation Demo **[3 Marks]**	Implementation is flawless, demonstrating a deep understanding of the project's technical aspects. **3 Marks**	Implementation is mostly successful, with minor issues or areas for improvement. **2 Marks**	Implementation has noticeable shortcomings or errors that require attention. **1 Marks**	No implementation demo provided. **0 Marks**
Subject Knowledge [VIVA] **[4 Marks]**	Demonstrates a profound understanding of the subject matter, answering questions with clarity and depth. **4 Marks**	Answers questions accurately and confidently, with minor gaps in understanding. **3 Marks**	Provides satisfactory answers but lacks depth or clarity in some areas. **2 Marks**	Unable to assess subject knowledge due to incomplete or missing VIVA. **0-1 Marks**
Report Writing **(Similarity/plagiarism must be less than 25 % for acceptance) [4 Marks]**	Report is well-written, organized, and thoroughly covers all necessary aspects with clarity. **4Marks**	Report is mostly clear and well-structured, with minor areas for improvement. **3 Marks**	Report has noticeable deficiencies in clarity, organization, or content. **2 Marks**	No report provided. **0-1 Marks**

Table 3: Project Evaluation Components Overview

S. No	**Components**	**Marks**
1	**Mid Review** Project: Title, Input, and Outputs, Dataset (1 Marks) Preliminarily results (2 mark) Future Plan (1 marks)	5 marks

Signature of the Course Faculty